Final Reproducible Pipeline Specification

Early clinical-domain prediction of visit-level 30-day mortality in acute care (CHoRUS development analysis and MIMIC-IV independent replication)

> **Verdict: YES, conditionally**
> This specification is sufficient to build a reusable and auditable pipeline for the paper. The reproducibility task is fulfilled only after the code, frozen configurations, synthetic tests, environment lockfile, manifests, and public-safe example run are released. The restricted CHoRUS data cannot be redistributed, so outside users can reproduce the method, not the exact CHoRUS records. Credentialed MIMIC-IV users can reproduce the replication if the same database version and configuration are available.

Exactly what the pipeline executes

- One dataset at a time: discover and confirm source mappings; freeze one acute-care cohort, predictor window, baseline features, and mortality outcome.
- Five patient-grouped folds: every patient appears in one validation fold only, and the same frozen folds are used by every model and feature matrix.
- Training-fold concept selection: within each fold, rank and select the 50 most prevalent eligible measurement, medication, and procedure concepts using training visits only; apply those concepts to both the training and held-out visits.
- Eight matrices and four models: evaluate baseline, three single-domain additions, three pairwise additions, and all domains using logistic regression, random forest, gradient boosting, and LightGBM.
- Out-of-fold evaluation: fit 8 x 4 x 5 = 160 fold models per dataset. Every visit receives exactly 32 OOF probabilities (one for each matrix-model combination).
- Numerical outputs only: save performance, confidence intervals, paired comparisons, ROC coordinates, calibration coordinates, clinical operating points, top-10% risk results, and decision-curve coordinates. Generate no plots.

> **Terminology correction**
> The pipeline does not create "21 out-of-fold features." It creates visit-level predictors from first-24-hour records and one held-out probability per visit for each model-matrix combination. Across the full design that is 32 OOF probabilities per visit.

Critical alignment with the manuscript

This final specification mandates concept selection inside each training fold. If the current manuscript results used one top-50 list selected across the complete cohort, all performance, calibration, clinical-utility, and feature-importance results must be rerun and the Methods updated. Do not mix global and fold-specific selection in one paper.

1. Pipeline architecture

Stage / file	Executes	Required output
01_discover_source_schema.py	Read-only metadata discovery for patient, encounter, diagnosis, death, demographics, dictionaries, and source-system signature. No patient rows.	source_mapping_suggestions.yaml; discovery_report.md
02_build_base_acute_care_cohort.py	Apply the confirmed mapping; freeze eligibility, landmark, predictor window, outcome, baseline variables, row order, and cohort checksum.	base_acute_care_cohort.parquet; baseline_X.parquet; cohort_manifest.json; attrition.csv
03_discover_clinical_domains.py	Discover and confirm measurement, medication, and procedure event tables, concept fields, timestamps, units, semantics, dictionaries, and join strategy.	clinical_domain_mapping.yaml; mapping_validation.json; mapping hash
04_create_fold_assignments.py	Assign each patient to one of five frozen GroupKFold folds. Validate both outcome classes in all train and validation partitions.	fold_assignments_restricted.csv; public fold summary; fold hash
05_build_measurement_features.py	For each fold, select training-only top 50 usable numeric concepts and construct six features per concept for train and validation visits.	Per-fold selections, 300-column matrices, feature dictionary, audits, manifests
06_build_medication_features.py	For each fold, select training-only top 50 concepts and construct exposure/count plus four aggregate features.	Per-fold selections, 104-column matrices, feature dictionary, audits, manifests
07_build_procedure_features.py	For each fold, select training-only top 50 concepts and construct exposure/count plus three aggregate features.	Per-fold selections, 103-column matrices, feature dictionary, audits, manifests
08_run_domain_analysis.py	Assemble eight fold-specific matrices, fit four frozen models, generate OOF probabilities, select one model per matrix, and calculate all analyses.	Master CSVs, restricted OOF predictions, per-matrix outputs, run manifest

Single source of truth

- One shared modeling_evaluation_utils.py implements folds, preprocessing, metrics, calibration, bootstrap, decision curves, and output schemas.
- Script 8 evaluates all eight matrices, including baseline. Do not maintain a second baseline evaluation engine.
- All choices come from versioned YAML/JSON configuration. Scripts must not infer new clinical definitions during execution.
- A top-level run_pipeline.py calls stages in order and stops at the first failed validation.

Execution order

Discover -> user confirms mappings -> freeze cohort -> freeze folds -> build fold-specific domain features -> fit models -> aggregate OOF results -> validate outputs. Discovery must never silently become analysis.

2. Frozen cohort and predictor contract

Item	Exact rule
Unit	One eligible parent acute-care visit; deterministic cohort_visit_number.
Population	Adult, acute, non-elective hospital/observation visits as prespecified in dataset configuration.
Landmark	visit_start_datetime + 24 hours.
Predictor window	[visit start, min(visit end when known, landmark)); the end is exclusive.
Early deaths	Exclude death on or before the landmark.
Outcome	1 only for death after the landmark and on/before 30 days from visit start; otherwise 0 with verified 30-day follow-up.
Short visits	Retain if the paper prespecifies them; predictor collection ends at the visit end. Record a short-visit flag for audit only.
Charlson	Non-age-adjusted score from diagnoses on prior admissions only, within 365 days before the index start; exclude index-visit billing diagnoses.
Baseline predictors	Age, sex, race, ethnicity, acute visit type, prior visit count, prior acute visit count, prior-visit indicator, and prior-only Charlson score.
Forbidden predictors	IDs, timestamps, outcome/death fields, discharge disposition, length of stay, post-landmark data, predictions, and derived future-risk fields.

Paper-specific configuration checks

Dataset	Visits	Patients	Deaths	Use
CHoRUS	22,098	5,892	1,004	Development analysis
MIMIC-IV	23,000	10,009	819	Independent replication

Store these as expected counts in the paper run configurations, not in reusable source logic. A mismatch stops the paper reproduction run and produces an attrition comparison. A new database run may use different counts under a new cohort version.

Cohort invariants

- The cohort is built once and never reconstructed by feature or model scripts.
- Every downstream table contains all and only the frozen cohort visits in the same deterministic order.
- All clinical events must link to exactly one cohort visit by confirmed visit ID, confirmed bridge, or patient plus time. Ambiguous links fail.
- Every input, mapping, configuration, row order, and output is checksummed and recorded in a manifest.
- CHoRUS and MIMIC-IV are never pooled. They use the same code and evaluation rules but separate mappings, concepts, folds, models, and outputs.

Interpretation of replication

MIMIC-IV is an independent replication of the domain-ranking pattern. It is not external validation of a CHoRUS-trained model, because each dataset selects concepts and trains models independently.

3. Fold-specific clinical features

For fold k, use only the other four folds to rank concepts. Select the 50 concepts with the greatest number of training visits containing at least one qualifying record; break ties by normalized concept key. Then construct the same selected columns for both the training visits and fold k validation visits. Outcomes are never used for ranking.

Domain	Selection and feature definition	Count
Measurements	50 training-selected usable numeric concepts. Per visit-concept: mean, max, min, sample SD (ddof=1; SD=0 for one value), valid-value count, and missingness indicator. With no valid value: summaries missing, count=0, missingness=1.	50 x 6 = 300
Medications	50 training-selected concepts. Per concept: qualifying-record exposure and count. Add any_drug_24h, unique_drug_count_24h, repeat_drug_exposure_count_24h, and time_to_first_drug_in_hours. Aggregates use all qualifying concepts, not only the top 50.	50 x 2 + 4 = 104
Procedures	50 training-selected concepts. Per concept: qualifying-record exposure and count. Add any_procedure_24h, unique_procedure_count_24h, and procedure_count_total_24h. Aggregates use all qualifying concepts.	50 x 2 + 3 = 103

Required feature rules

- Semantics: describe what the source records actually represent. Orders are not administrations; claims or orders are not necessarily performed procedures.
- Measurements: do not coerce categorical results into numbers or combine incompatible units. Unit and conversion rules must be clinically confirmed before modeling or learned from training data only.
- Missingness: distinguish no record from unusable records in the audit. Continuous measurement summaries remain missing until training-fold median imputation.
- Zero-event visits: remain in every matrix. Medication/procedure exposure and count fields are zero; time to first drug is missing when no drug exists.
- Selection records: save rank, training prevalence, selected concept identifiers/names, source semantics, units, fold, and selection hash. The selected vocabulary may differ across folds and datasets.
- Feature importance: because folds can use different concepts, report concept selection frequency and fold-level importance. A full-data explanatory fit may be added but must not be used for performance estimation.

No validation-fold leakage

Validation visits may supply their own first-24-hour values for the concepts already chosen by the training folds. They must not influence concept ranking, unit choice, imputation, encoding, scaling, variance filtering, threshold selection, or model fitting.

4. Feature matrices, models, and OOF predictions

Matrix ID	Components
baseline	Baseline
baseline_measurements	Baseline + measurements
baseline_medications	Baseline + medications
baseline_procedures	Baseline + procedures
baseline_measurements_medications	Baseline + measurements + medications
baseline_measurements_procedures	Baseline + measurements + procedures
baseline_medications_procedures	Baseline + medications + procedures
baseline_all_domains	Baseline + measurements + medications + procedures

Frozen models

Model	Exact estimator settings	Fold-safe preprocessing
Logistic regression	max_iter=1000, random_state=42	Median imputation; one-hot encoding; StandardScaler
Random forest	n_estimators=300, random_state=42	Median imputation; one-hot encoding
Gradient boosting	GradientBoostingClassifier(random_state=42)	Median imputation; one-hot encoding
LightGBM	n_estimators=500; learning_rate=0.03; num_leaves=31; max_depth=-1; subsample=0.8; subsample_freq=1; colsample_bytree=0.8; random_state=42	Median imputation; one-hot encoding

OOF algorithm

- Initialize one empty probability vector for every matrix-model combination.
- For fold k, load its training-selected feature columns; fit all preprocessing and the model on the other four folds only; predict class-1 probability for fold k only.
- Repeat for five folds, then assert exactly one numeric probability in [0,1] per visit and combination.
- Reuse the same patient folds for all matrices and models. Do not add a second train/test split, tune hyperparameters, use SMOTE, or use in-sample predictions.
- Save patient-level OOF predictions only in restricted storage. Public outputs contain aggregate values and non-identifying row keys only when safe.

Model selection statement

Select the best model separately for each matrix by pooled OOF AUPRC, then AUROC, then lower Brier score, then a frozen model order. Because selection and evaluation use the same OOF predictions, describe the result as an internally selected model; do not call it an independent test estimate.

5. Exact analyses and outputs

Analysis	Exact computation	Primary file
Performance	Fold-level, mean +/- SD, and pooled OOF AUROC, average precision (AUPRC), Brier, log loss; threshold 0.5 confusion counts, accuracy, balanced accuracy, PPV, NPV, sensitivity, specificity, and F1.	selected_model_performance_table.csv
Confidence intervals	2,000 percentile bootstrap resamples at the patient level; include all visits for sampled patients; use identical resamples for paired comparisons; report invalid replicates.	all_metric_confidence_intervals.csv
ROC and 90% specificity	Save complete ROC coordinates. Among points with specificity >=0.90, choose greatest sensitivity; tie by specificity closest to 0.90, then highest threshold.	selected_models_roc_coordinates.csv; selected_models_sensitivity_at_90_specificity.csv
Top 10% risk	Rank OOF risk descending; tie by cohort_visit_number; flag exactly ceil(0.10N). Report deaths captured, PPV, flagged per 100, cutoff ties, and enrichment = top-10% PPV / prevalence.	selected_models_top_10_percent_risk_analysis.csv
Calibration	Unmodified OOF probabilities; Brier, intercept, slope, mean prediction, prevalence, ECE; 10 quantile bins with counts, mean risk, observed rate, and event-rate CI.	selected_models_calibration_summary.csv; selected_models_calibration_coordinates.csv
Decision curve	Thresholds 0.01-0.50 by 0.01. NB(model)=TP/N-FP/N*(pt/(1-pt)); include treat-all and treat-none and patient-bootstrap CI.	selected_models_decision_curve_coordinates.csv
Paired comparisons	Patient-paired bootstrap differences for baseline additions, pairwise complements, and all-domain versus each pairwise matrix. Define Brier improvement as comparator minus expanded.	prespecified_paired_matrix_comparisons.csv

Three manuscript-ready tables

- Performance: selected model, AUROC and CI, AUPRC and CI, Brier and CI, plus threshold-0.5 metrics.
- Clinical utility: sensitivity at 90% specificity, achieved specificity, PPV and flagged per 100 at that threshold, deaths captured and PPV in the highest-risk 10%, and enrichment.
- Calibration: Brier, intercept, slope, mean predicted risk, observed prevalence, ECE, and requested/actual bin counts.

Analytical CSVs retain full precision. Separate display tables may round values. Plotting is outside this pipeline and must read the saved coordinate files without recomputing statistics.

6. Reproducibility and acceptance gates

Repository item	Required content
README	Data-access instructions, exact stage order, one command per dataset, expected outputs, restricted/public boundary, and Methods-to-config crosswalk.
Configuration	Dataset cohort config; confirmed source/domain mappings; model config; evaluation config; seeds; hashes; paper expected counts.
Environment	Pinned Python and package versions in a lockfile/container; record pandas, NumPy, scikit-learn, LightGBM, SciPy, and PyArrow versions.
Tests	Synthetic unit/integration tests for landmark boundaries, patient leakage, ambiguous joins, incompatible units, row loss/duplication, forbidden predictors, fold-specific selection, invalid probabilities, and deterministic reruns.
Manifests	Input/output checksums, Git commit, run timestamp, cohort/mapping/fold/selection hashes, feature names/counts, model settings, software versions, warnings, and failures.
Example	A public synthetic dataset and expected aggregate outputs that run end to end without protected data.

Mandatory failures

- Unconfirmed or changed mapping; input checksum/hash mismatch; missing source column/table; cohort count mismatch in a paper run.
- Event outside the predictor window; ambiguous event-to-visit assignment; incompatible measurement units; duplicate event/feature keys.
- Any visit added, lost, duplicated, or reordered without an audited one-to-one reconciliation; patient assigned to multiple folds.
- Concept ranking uses validation visits; preprocessing is fitted before the fold split; unexpected, identifying, temporal, outcome, discharge, or post-landmark field enters X.
- A visit receives zero or multiple OOF predictions; wrong positive class; invalid probabilities; differing folds across models/matrices; missing required output.
- Identical inputs, configs, seeds, code, and environment produce different analytical outputs.

Public versus restricted artifacts

Public repository	Restricted local storage
Source code; synthetic tests/data; configs without secrets; source/domain mappings if non-sensitive; data dictionaries; aggregate counts; coordinate and results tables cleared for release; manifests without patient values.	Raw clinical data; credentials; patient/source visit identifiers; dates/times; event-level audit samples; patient-to-fold assignments; patient-level OOF predictions; any small-cell outputs prohibited by the data-use agreement.

What this pipeline does not establish

Reproducible code does not by itself prove transportability, causal effects, or deployment readiness. The pooled-OOF 90%-specificity threshold is descriptive, not a prospectively validated operating threshold. Nested threshold selection or a held-out test set is needed for a deployment claim.

7. Definition of done for this paper

#	Acceptance criterion
1	Both datasets use the same committed code and evaluation configuration, but separate confirmed mappings, folds, selections, models, and outputs.
2	The CHoRUS and MIMIC-IV paper configurations reproduce the frozen cohort counts and outcome definitions shown in this document.
3	All eight matrices are evaluated with all four models using the same five patient-grouped folds.
4	Top-50 clinical concepts are ranked inside each training fold; every per-fold selection list and hash is retained.
5	Every visit receives exactly one OOF probability per matrix-model combination and no training prediction enters evaluation.
6	All imputation, encoding, scaling, and other learned preprocessing are fitted on training visits only.
7	Selected-model performance, clinical utility, calibration, decision curves, and patient-bootstrap CIs are generated from OOF predictions.
8	Prespecified paired domain-increment comparisons use identical patient bootstrap samples.
9	Leakage, row-integrity, fold-integrity, event-window, unit, and determinism tests pass; mandatory failures stop the run.
10	Code, configs, lockfile/container, synthetic example, README, output schemas, and public-safe aggregate outputs are released; restricted data remain local.
11	The manuscript Methods, supplement, table values, and figure coordinates match the final configs and run manifests.
12	A clean-machine rerun of the synthetic example succeeds; a credentialed MIMIC-IV rerun reproduces the replication within documented software/data-version constraints.

Final assessment

The revised specification fulfills the reproducible-pipeline task at the design level. It is substantially more concise than the original 189-page document and removes the main ambiguity: feature concepts are selected in the training portion of each fold, then applied to that fold's held-out visits. The task is not complete merely because this PDF exists; completion requires a tested implementation and release package satisfying all 12 acceptance criteria above.

Recommended top-level commands

Purpose	Command contract
Validate code	python run_pipeline.py --synthetic-test
CHoRUS run	python run_pipeline.py --dataset chorus --config configs/chorus_paper.yaml
MIMIC-IV run	python run_pipeline.py --dataset mimiciv --config configs/mimiciv_paper.yaml
Verify release	python run_pipeline.py --verify-manifest outputs//run_manifest.json

Version: 1.0 | Date: 27 July 2026 | Scope: implementation instructions for the CHoRUS/MIMIC-IV clinical-domain mortality-prediction paper.